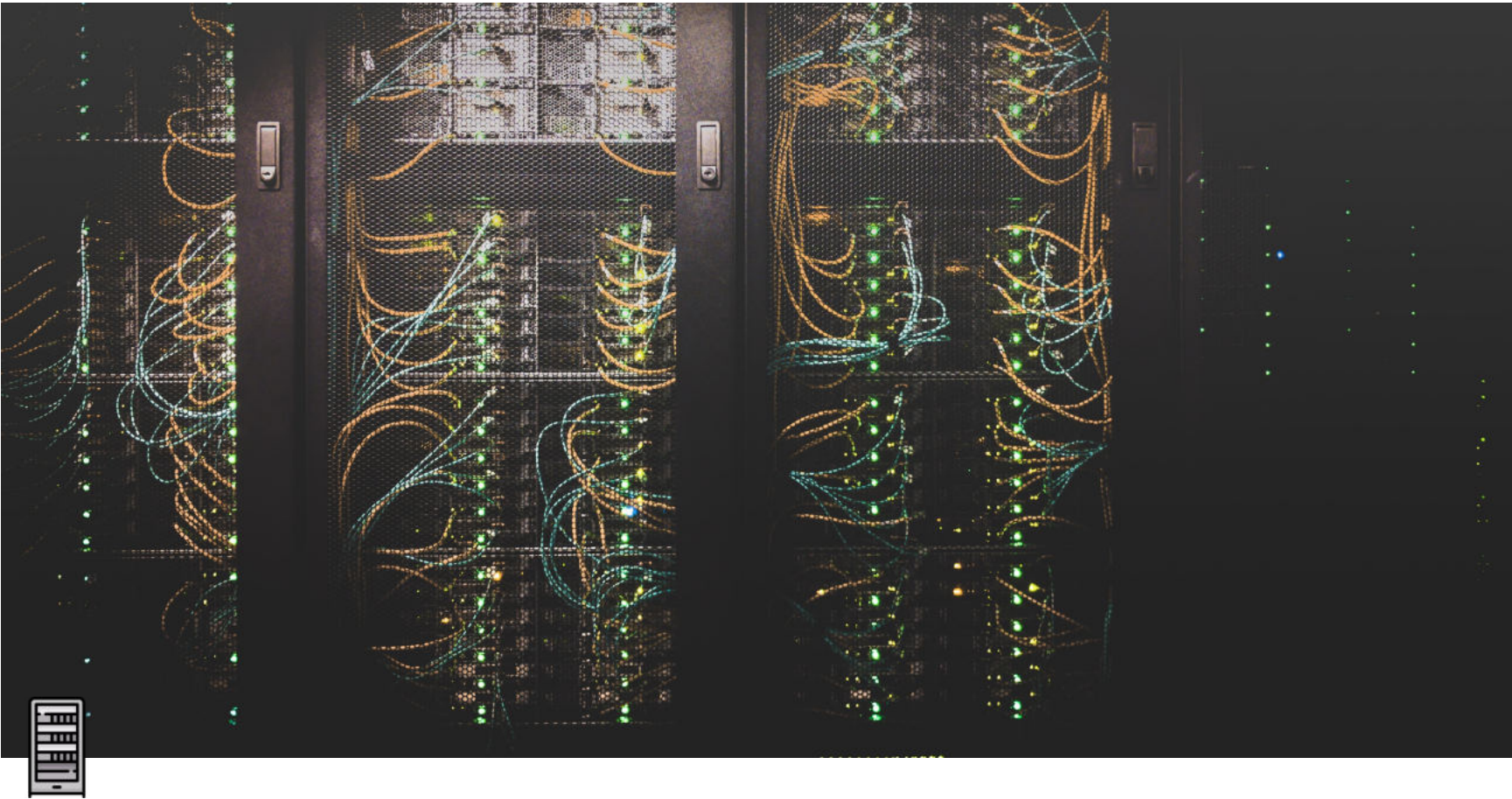


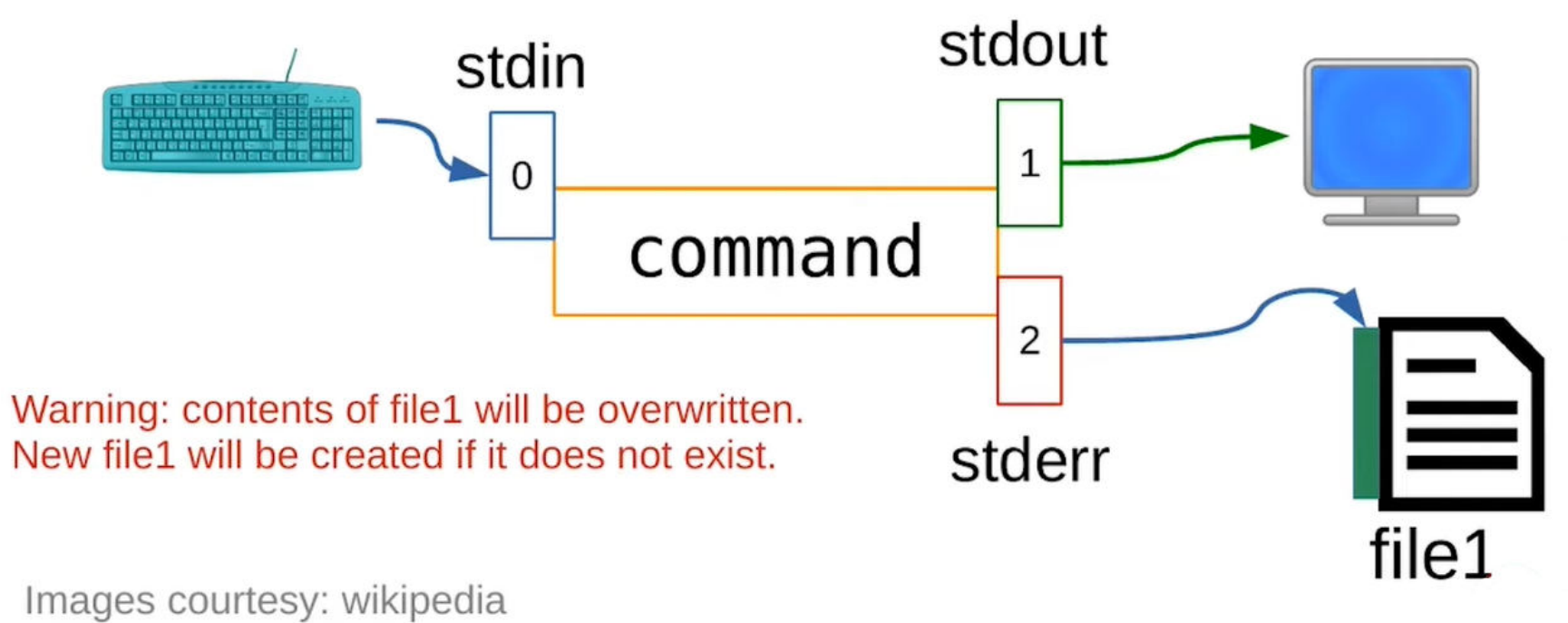


Redirections

Type	Lecture
Date	@January 9, 2022
Lecture #	2
Lecture URL	https://youtu.be/BBh69kH_G_Y
Notion URL	https://21f1003586.notion.site/Redirections-734673f36f21448f99de25ccb092c8d4
Week #	3

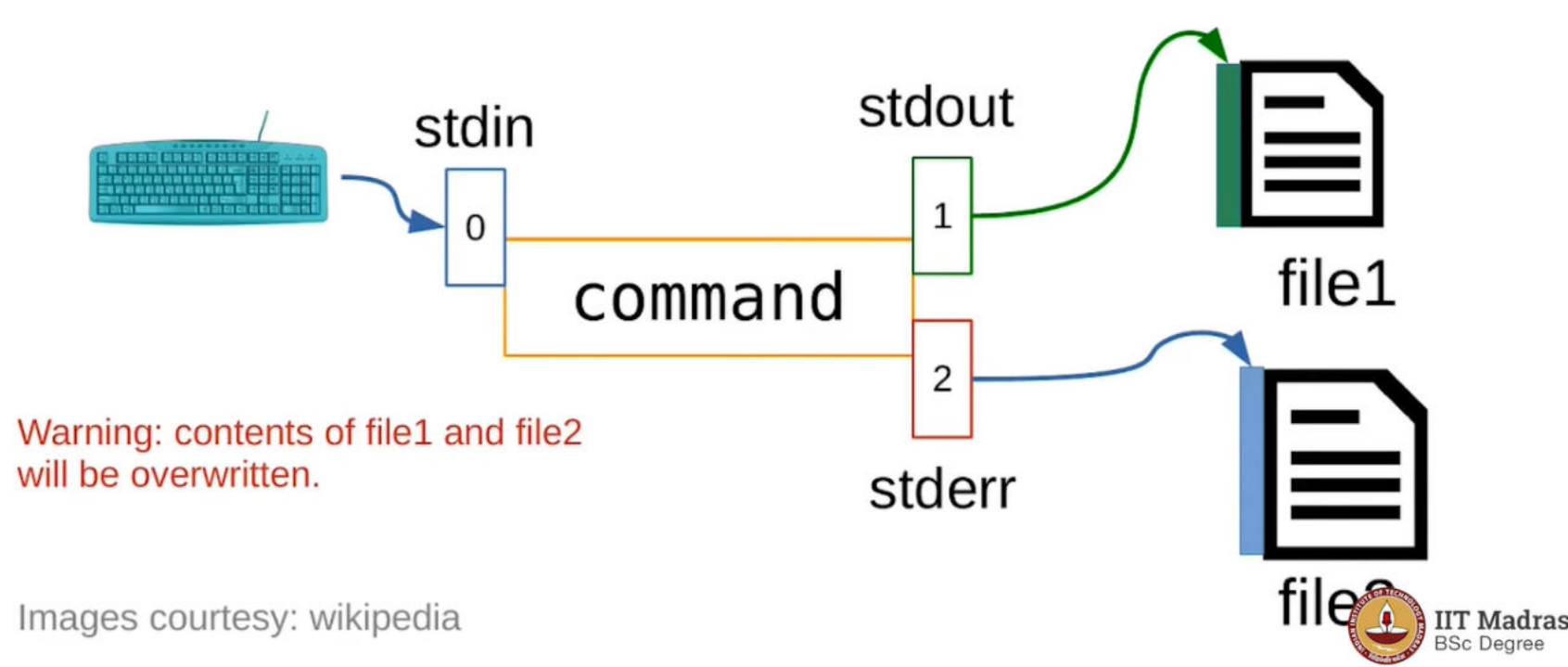
command 2> file1

- Redirect the output of the `command` to `stdout` , which is the display in this case
- Redirect the error of the `command` to `file1`



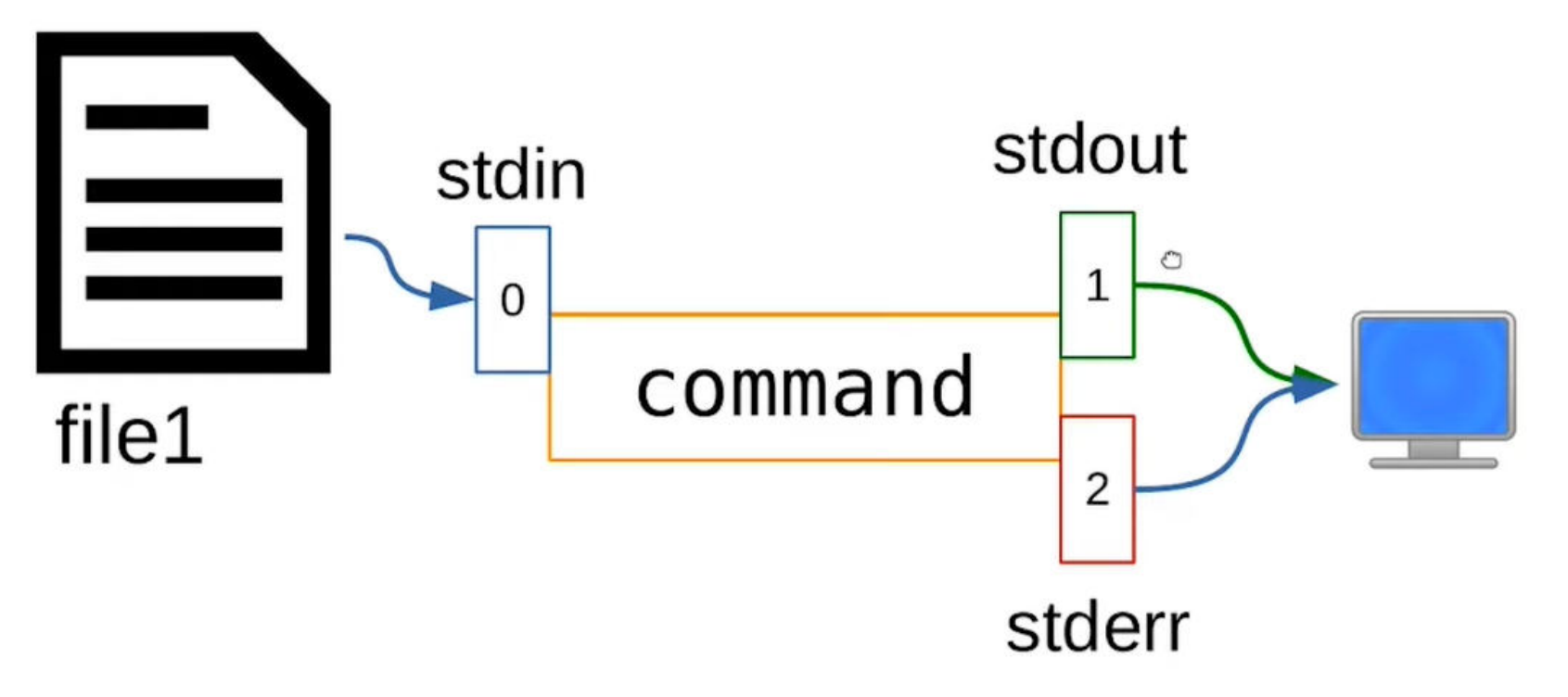
command > file1 2> file2

- Redirect the output of the `command` to the `stdout` , i.e. `file1`
- Redirect the error of the `command` to `stderr` , i.e. `file2`



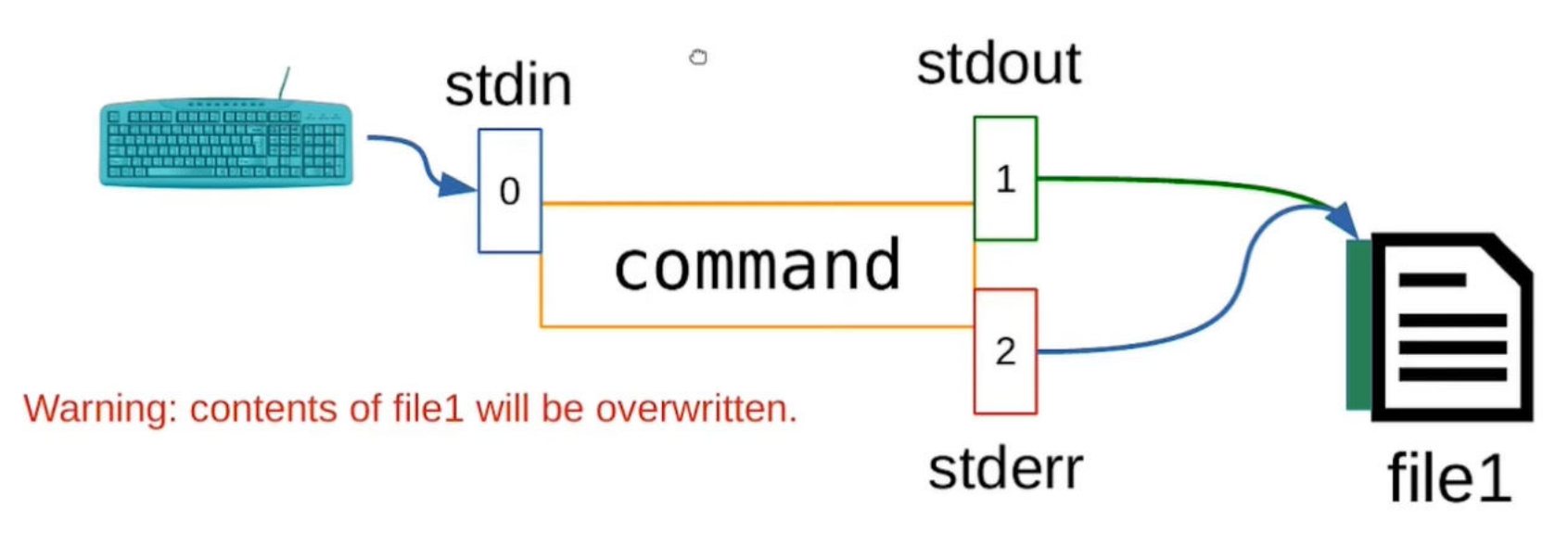
```
command < file1
```

- Any `command` which takes input from the keyboard, now takes input from `file1`



```
command > file1 2>&1
```

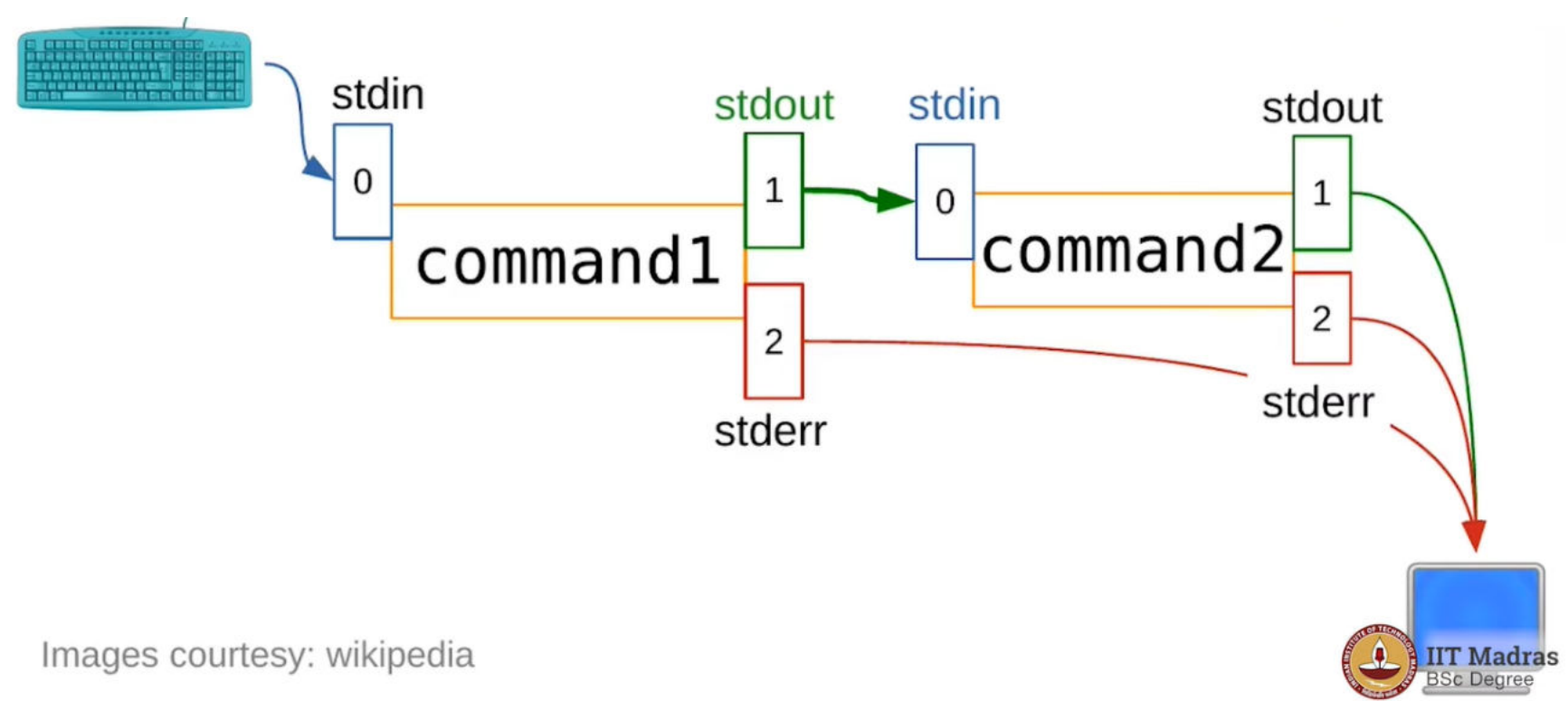
- The output of `command` is written to `file1`
- The error is redirected to stream 1, which is `stdout`



```
command1 | command2 → pipe operator |
```

- The output of `command1` is sent to `command2` as input

- By default, the `stderr` will output to the display



Images courtesy: wikipedia

Count the number of files in the directory `/usr/bin`

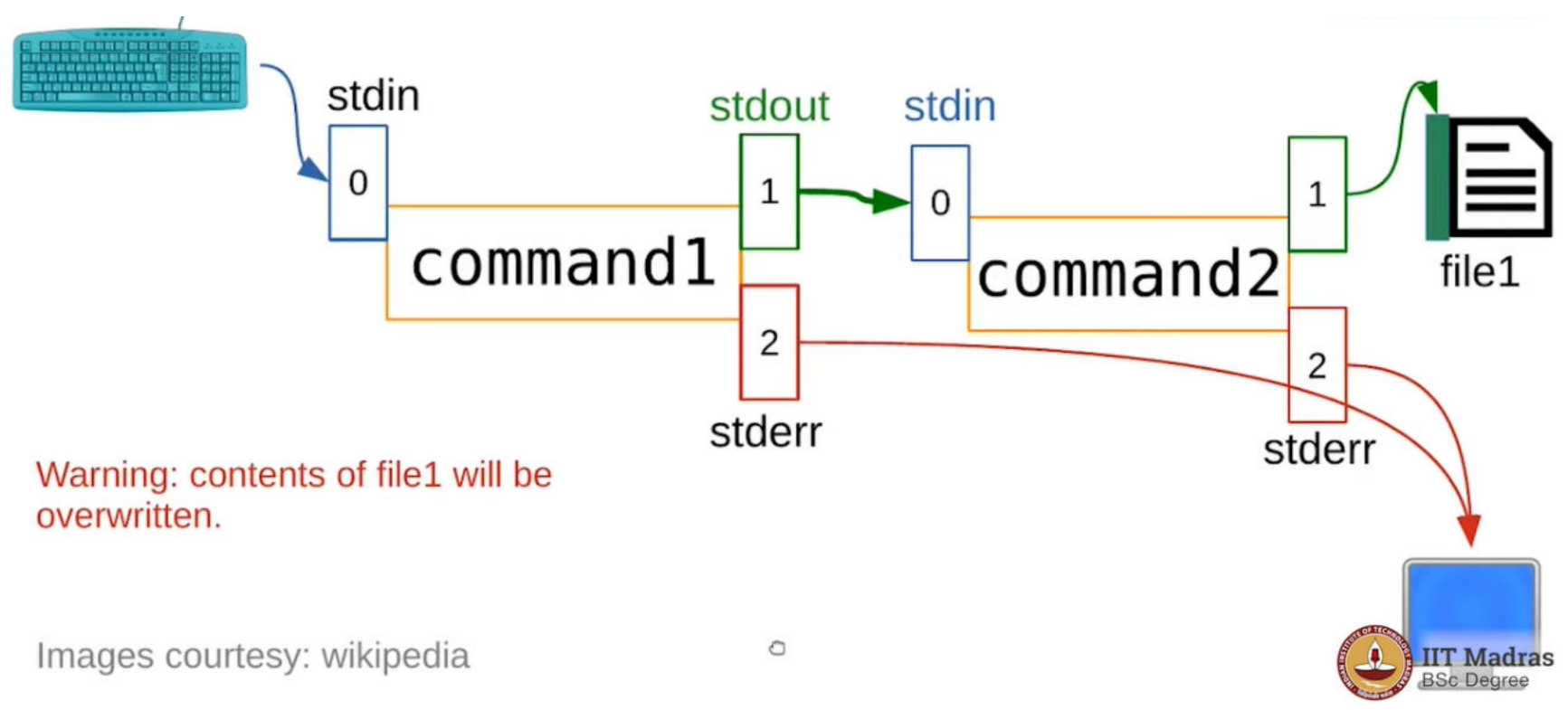
```
kashif@Zen:~$ ls /usr/bin/ | wc -l
1603
```

List the files of `/usr/bin` directory, but use the `less` command to scroll at ease

```
ls /usr/bin | less
```

```
command1 | command2 > file1
```

- The `stdout` of `command1` is mapped to `stdin` of `command2`
- The `stdout` of `command2` is written to `file1`
- The `stderr` is output to the display



Images courtesy: wikipedia

```
/dev/null
```

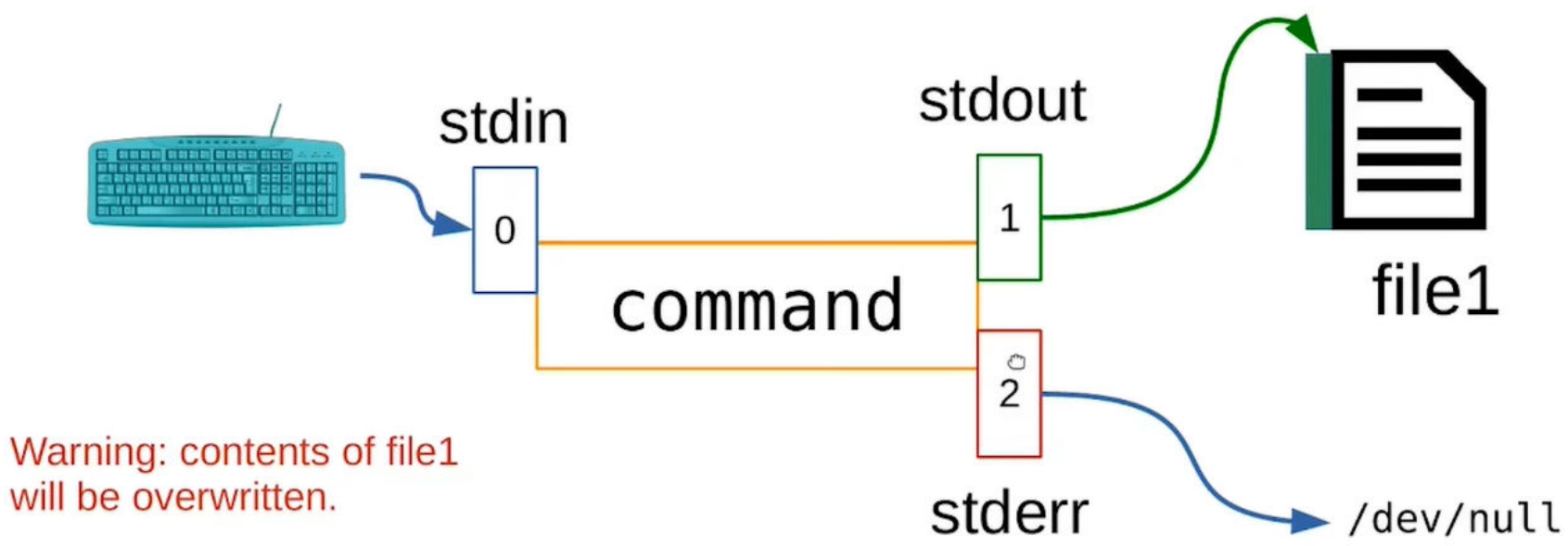
- A sink for output to be discarded
- Use `→` silent and clean scripts

So, a typical usage looks like ...

```
command > file1 2> /dev/null
```

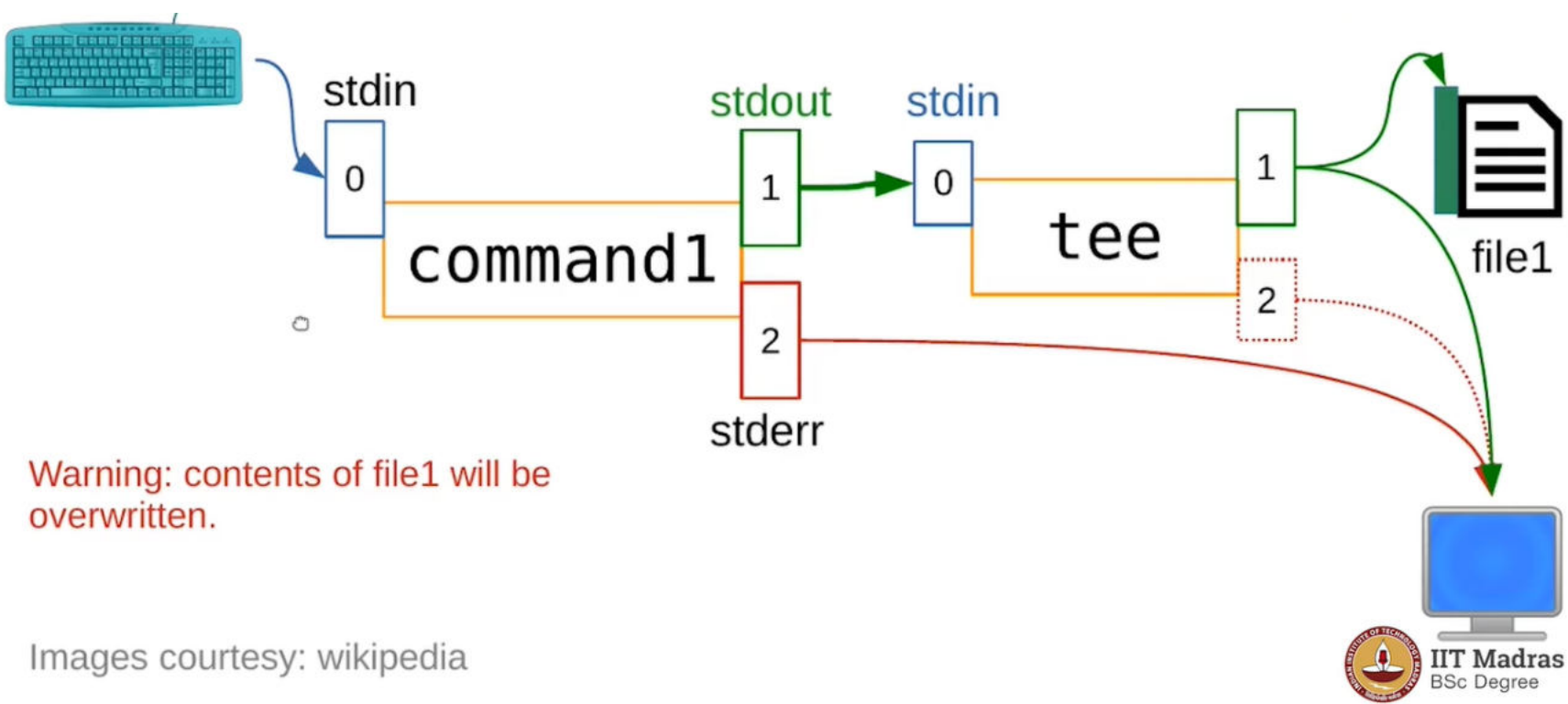
- The output of `command` is written to `file1`

- The `stderr` is written to `/dev/null`, ~~which gets warped to another dimension~~



```
command1 | tee file1
```

- The `tee` command splits the output into 2 streams, one stream is written to the `file1` another, one to the display
 - This command can write to multiple files as well



`diff` command

- This command compares files line-by-line

Usage

```
diff file1 file2
```